

# SOOHYUN CHOI

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning  
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## RESEARCH INTERESTS

Reinforcement learning and sequential decision making, with emphasis on multi-step policy improvement, analysis of tau regimes, offline goal-conditioned RL, and generative long-horizon state-space planning.

## EDUCATION

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| <b>Hanyang University</b><br><i>Integrated M.S./Ph.D. Program in Electronic Engineering</i><br>Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong | 2024-Present |
| <b>Hanyang University</b><br><i>B.S. in Electronic Engineering</i>                                                                                                           | 2018-2024    |

## MANUSCRIPT & SELECTED RESEARCH

**S. Choi, S. Cho, and S. Hong**, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning." *Under Review, AAAI 2027*.

- Developed explicit state-space bridges, inverse-dynamics action chunks, and execution-time replanning for long-horizon offline GCRL; evaluated on OGBench navigation and manipulation tasks.

**Multi-step Proximal Policy Improvement** 2026-Present

- Analyze how proximal step size tau governs recursively re-centered policy refinement, using sweeps and mathematical displacement diagnostics to characterize local-flow, nonlocal finite-step, and over-refinement regimes.

**PathFlower - Goal-Conditioned State Flows** Current

- Extend PathBridger from an explicit subgoal and bridge toward final-goal-conditioned state flows that generate useful intermediate states for long-horizon planning.

**SK hynix-Hanyang University Industry-Academic Research Project** 2024-2025

- Contributed to machine-learning- and transfer-function-based modeling for circuit-interconnect waveform and delay estimation, including model validation and industry technical exchanges.

## HONORS & FUNDING

- Outstanding Graduation Project Award - KRW 2,000,000 award
- Excellence Award, College of Engineering Capstone Design - KRW 1,000,000 award
- BK Research Assistantship (BK-RA), Hanyang University - 2025

## SELECTED GRADUATE COURSEWORK

**Learning & Control:** Machine Learning; Robot Learning; Advanced Optimal Control Theory

**Probability & Analysis:** Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I